

6. Orthogonality and Least Squares

6.1. Inner Product, Length, and Orthogonality

Inner Product (Dot Product)

If $\vec{u}$ and $\vec{v}$ are vectors in $\mathbb{R}^n$, the number $\vec{u} \cdot \vec{v}$ is called the inner product / dot product of $\vec{u}$ and $\vec{v}$.

Theorem 1

- a. $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$
- b. $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$
- c. $(c\vec{u}) \cdot \vec{v} = c(\vec{u} \cdot \vec{v}) = \vec{u} \cdot (c\vec{v})$
- d. $\vec{u} \cdot \vec{u} \geq 0$, and $\vec{u} \cdot \vec{u} = 0$ if and only if $\vec{u} = \vec{0}$
 $(c_1\vec{u}_1 \dots c_p\vec{u}_p) \cdot \vec{w} = c_1(\vec{u}_1 \cdot \vec{w}) \dots c_p(\vec{u}_p \cdot \vec{w})$

Definition: Length/Norm

The length (or norm) of a vector $\vec{v}$ in $\mathbb{R}^n$ is the nonnegative scalar $\|\vec{v}\|$, defined by:

$$\|\vec{v}\| = \sqrt{\vec{v} \cdot \vec{v}} = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

and then

$$\|\vec{v}\|^2 = \vec{v} \cdot \vec{v}$$

$$\|c\vec{v}\| = |c|\|\vec{v}\|$$

Definition: Unit Vector and Normalize

A vector whose length is 1 is a unit vector. The process of creating a unit vector from $\vec{v}$ is normalizing $\vec{v}$, and the unit vector is in the same direction as $\vec{v}$.

$$\vec{u} = \vec{v} / \|\vec{v}\|$$

Definition: Distance

For $\vec{u}$ and $\vec{v}$ in $\mathbb{R}^n$, the distance between $\vec{u}$ and $\vec{v}$, written as $\text{dist}(\vec{u}, \vec{v})$, is the length of the vector $\vec{u} - \vec{v}$:

$$\text{dist}(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\|$$

Definition: Orthogonal

Two vectors $\vec{u}$ and $\vec{v}$ in $\mathbb{R}^n$ are said to be orthogonal (or perpendicular) if $\vec{u} \cdot \vec{v} = 0$.

Theorem 2: The Pythagorean Theorem

Two vectors $\vec{u}$ and $\vec{v}$ are orthogonal if and only if:

$$\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2$$

Orthogonal Complement

If a vector $\vec{z}$ is orthogonal to every vector in a subspace W of $\mathbb{R}^n$, then $\vec{z}$ is said to be orthogonal to W . The set of all such vectors $\vec{z}$ is called the orthogonal complement of W , denoted by $W^\perp$:

$$W^\perp = \{\vec{z} \in \mathbb{R}^n \mid \vec{z} \cdot \vec{w} = 0 \text{ for all } \vec{w} \in W\}$$

Properties of Orthogonal Complement

1. $W^\perp$ is a subspace of $\mathbb{R}^n$.
2. a vector $\vec{x}$ is in $W^\perp$ if and only if $\vec{x}$ is orthogonal to every vector in a set that spans W .

Theorem 3

Let A be an $m \times n$ matrix. The orthogonal complement of the row space of A is the null space of A , and the orthogonal complement of the column space of A is the null space of $A^\top$:

$$(\text{Row } A)^\perp = \text{Null } A$$

$$(\text{Col } A)^\perp = \text{Null } A^\top$$

6.2. Orthogonal Sets

Definition: Orthogonal Set

A set of vectors $\{\vec{u}_1, \dots, \vec{u}_p\}$ in $\mathbb{R}^n$ is an orthogonal set if each pair of distinct vectors from the set is orthogonal. That is, $\vec{u}_i \cdot \vec{u}_j = 0$ when $i \neq j$.

Theorem 4

If $S = \{\vec{u}_1, \dots, \vec{u}_p\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$, then S is linearly independent and hence is a basis for the subspace spanned by S .

Definition: Orthogonal Basis

An orthogonal basis for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set.

Theorem 5

Let $\{\vec{u}_1, \dots, \vec{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $\vec{y}$ in W , the weights in the linear combination $\vec{y} = c_1\vec{u}_1 + c_2\vec{u}_2 + \dots + c_p\vec{u}_p$ are given by:

$$c_i = \frac{\vec{y} \cdot \vec{u}_i}{\vec{u}_i \cdot \vec{u}_i} \quad (i = 1, 2, \dots, p)$$

Orthogonal Projection onto a Vector

Given a nonzero vector $\vec{u}$ in $\mathbb{R}^n$, consider decomposing a vector $\vec{y}$ in $\mathbb{R}^n$ into a sum of two vectors: one a multiple of $\vec{u}$, and the other orthogonal to $\vec{u}$:

$$\vec{y} = \hat{\vec{y}} + \vec{z}$$

where $\hat{\vec{y}} = \alpha\vec{u}$ (a scalar multiple of $\vec{u}$) and $\vec{z} = \vec{y} - \hat{\vec{y}}$ (orthogonal to $\vec{u}$).

Condition for Orthogonality

$\vec{z} = \vec{y} - \alpha\vec{u}$ is orthogonal to $\vec{u}$ if and only if:

$$0 = (\vec{y} - \alpha\vec{u}) \cdot \vec{u} = \vec{y} \cdot \vec{u} - \alpha(\vec{u} \cdot \vec{u})$$

Solving for α gives:

$$\alpha = \frac{\vec{y} \cdot \vec{u}}{\vec{u} \cdot \vec{u}}$$

Definitions

- The vector $\hat{\vec{y}} = \frac{\vec{y} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{u}$ is the orthogonal projection of $\vec{y}$ onto $\vec{u}$.
- The vector $\vec{z} = \vec{y} - \hat{\vec{y}}$ is the component of $\vec{y}$ orthogonal to $\vec{u}$.

Sometimes $\hat{\vec{y}}$ is denoted by $\text{proj}_L \vec{y}$ and is called the orthogonal projection of $\vec{y}$ onto L . That is

$$\hat{\vec{y}} = \text{proj}_L \vec{y} = \frac{\vec{y} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{u}$$

Definition: Orthonormal Set

A set $\{\vec{u}_1, \dots, \vec{u}_p\}$ is an orthonormal set if it is an orthogonal set of unit vectors (i.e., $\|\vec{u}_i\| = 1$ for all i , and $\vec{u}_i \cdot \vec{u}_j = 0$ for $i \neq j$). If W is the subspace spanned by such a set, then the set is an orthonormal basis for W .

Theorem 6

An $m \times n$ matrix U has orthonormal columns if and only if $U^\top U = I_n$ (where I_n is the $n \times n$ identity matrix).

Theorem 7

Let U be an $m \times n$ matrix with orthonormal columns, and let $\vec{x}$ and $\vec{y}$ be in $\mathbb{R}^n$. Then:

- $\|U\vec{x}\| = \|\vec{x}\|$ (preserves length)
- $(U\vec{x}) \cdot (U\vec{y}) = \vec{x} \cdot \vec{y}$ (preserves inner product)
- $(U\vec{x}) \cdot (U\vec{y}) = 0$ if and only if $\vec{x} \cdot \vec{y} = 0$ (preserves orthogonality)

Definition: Orthogonal Matrix

An $n \times n$ matrix U is an orthogonal matrix if its columns are orthonormal (equivalently, $U^\top U = I_n$ and $UU^\top = I_n$). This implies $U^\top = U^{-1}$ (the transpose is the inverse).

Properties of Orthogonal Matrices

- If U is an orthogonal matrix, then $U^{-1} = U^\top$.
- $\det(U) = 1$ or $\det(U) = -1$ (absolute value of determinant is 1).
- If A and B are orthogonal matrix, then AB is an orthogonal matrix.

Definition: Orthogonal Transformation

If U is an orthogonal matrix, then the linear transformation $\vec{y} = U\vec{x}$ is an orthogonal transformation. (Preserves length and orthogonality, by Theorem 7)

6.3. Orthogonal Projection

Theorem 8: The Orthogonal Decomposition Theorem

Let W be a subspace of $\mathbb{R}^n$. Then each $\vec{y}$ in $\mathbb{R}^n$ can be written uniquely in the form:

$$\vec{y} = \hat{\vec{y}} + \vec{z}$$

where $\hat{\vec{y}} \in W$ and $\vec{z} \in W^\perp$. If $\{\vec{u}_1, \dots, \vec{u}_p\}$ is an orthogonal basis of W , then:

$$\hat{\vec{y}} = \frac{\vec{y} \cdot \vec{u}_1}{\vec{u}_1 \cdot \vec{u}_1} \vec{u}_1 + \dots + \frac{\vec{y} \cdot \vec{u}_p}{\vec{u}_p \cdot \vec{u}_p} \vec{u}_p$$

$$\text{and } \vec{z} = \vec{y} - \hat{\vec{y}}.$$

Definition: Orthogonal Projection onto a Subspace

The vector $\hat{\vec{y}}$ in the orthogonal decomposition is called the orthogonal projection of $\vec{y}$ onto W , denoted $\text{proj}_W \vec{y}$.

If $\vec{y}$ is in $W = \text{Span}\{\vec{u}_1, \dots, \vec{u}_p\}$, then $\text{proj}_W \vec{y} = \vec{y}$.

Theorem 9: The Best Approximation Theorem

Let W be a subspace of $\mathbb{R}^n$, let $\vec{y}$ be any vector in $\mathbb{R}^n$, and let $\hat{\vec{y}}$ be the orthogonal projection of $\vec{y}$ onto W . Then $\hat{\vec{y}}$ is the closest point in W to $\vec{y}$ —that is:

$$\|\vec{y} - \hat{\vec{y}}\| < \|\vec{y} - \vec{w}\|$$

for all $\vec{w} \in W$ with $\vec{w} \neq \hat{\vec{y}}$. The vector $\hat{\vec{y}}$ is the best approximation to $\vec{y}$ by elements of W .

Projection Matrix

If W is a subspace of $\mathbb{R}^n$ with orthogonal basis $\{\vec{u}_1, \dots, \vec{u}_p\}$, then the orthogonal projection of $\vec{y}$ onto W can be written as $\text{proj}_W \vec{y} = P\vec{y}$, where P is the projection matrix:

$$P = \vec{u}_1 \vec{u}_1^\top + \vec{u}_2 \vec{u}_2^\top + \dots + \vec{u}_p \vec{u}_p^\top$$

If $U = [\vec{u}_1 \ \vec{u}_2 \ \dots \ \vec{u}_p]$ (an $n \times p$ matrix with columns as the orthogonal basis), then:

$$P = U(U^\top U)^{-1}U^\top = UU^\top$$

(If the basis is orthonormal, $U^\top U = I_p$, so $P = UU^\top$)

6.4. The Gram-Schmidt Process

Theorem 11: The Gram-Schmidt Process

Given a basis $\{\vec{x}_1, \dots, \vec{x}_p\}$ for a nonzero subspace W of $\mathbb{R}^n$, define:

$$\vec{v}_1 = \vec{x}_1$$

$$\vec{v}_2 = \vec{x}_2 - \frac{\vec{x}_2 \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1} \vec{v}_1$$

$$\vec{v}_3 = \vec{x}_3 - \frac{\vec{x}_3 \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1} \vec{v}_1 - \frac{\vec{x}_3 \cdot \vec{v}_2}{\vec{v}_2 \cdot \vec{v}_2} \vec{v}_2$$

$\vdots$

$$\vec{v}_p = \vec{x}_p - \frac{\vec{x}_p \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1} \vec{v}_1 - \frac{\vec{x}_p \cdot \vec{v}_2}{\vec{v}_2 \cdot \vec{v}_2} \vec{v}_2 - \dots - \frac{\vec{x}_p \cdot \vec{v}_{p-1}}{\vec{v}_{p-1} \cdot \vec{v}_{p-1}} \vec{v}_{p-1}$$

Then $\{\vec{v}_1, \dots, \vec{v}_p\}$ is an orthogonal basis for W , and $\text{Span}\{\vec{v}_1, \dots, \vec{v}_k\} = \text{Span}\{\vec{x}_1, \dots, \vec{x}_k\}$ for each $k = 1, 2, \dots, p$.

Theorem 12: The QR Factorization

If A is an $m \times n$ matrix with linearly independent columns, then A can be factored as $A = QR$, where:

- Q is an $m \times n$ matrix whose columns form an orthonormal basis for $\text{Col } A$.
- R is an $n \times n$ upper triangular invertible matrix with positive entries on its diagonal.

Relationships

$$R = Q^\top A$$

$$Q^\top Q = I_n$$

$$R_{ii} > 0 \quad (i = 1, 2, \dots, n)$$

6.5. Least-Squares Problems

Definition: Least-Squares Solution

Given an $m \times n$ matrix A and a vector $\vec{b}$ in $\mathbb{R}^m$, a least-squares solution of $A\vec{x} = \vec{b}$ is a vector $\hat{\vec{x}}$ in $\mathbb{R}^n$ such that:

$$\|A\hat{\vec{x}} - \vec{b}\| \leq \|A\vec{x} - \vec{b}\|$$

for all $\vec{x}$ in $\mathbb{R}^n$. The vector $A\hat{\vec{x}}$ is the orthogonal projection of $\vec{b}$ onto $\text{Col } A$, and $\|A\hat{\vec{x}} - \vec{b}\|$ is the least-squares error.

Theorem 13

The set of least-squares solutions of $A\vec{x} = \vec{b}$ coincides with the nonempty set of solutions of the normal equations:

$$A^\top A\vec{x} = A^\top \vec{b}$$

Theorem 14

Let A be an $m \times n$ matrix. The following are equivalent:

- The equation $A\vec{x} = \vec{b}$ has a unique least-squares solution for each $\vec{b}$ in $\mathbb{R}^m$.
- The columns of A are linearly independent.
- The matrix $A^\top A$ is invertible.

When these conditions hold, the unique least-squares solution is:

$$\hat{\vec{x}} = (A^\top A)^{-1} A^\top \vec{b}$$

The distance from $\vec{b}$ to $A\vec{x}$ is the least-squares error of this approximation.

Theorem 15

Given an $m \times n$ matrix A with linearly independent columns, let $A = QR$ be a QR factorization of A (as in Theorem 12). Then for each $\vec{b}$ in $\mathbb{R}^m$, the equation $A\vec{x} = \vec{b}$ has a unique least-squares solution given by:

$$\hat{\vec{x}} = R^{-1}Q^\top \vec{b}$$

Proof

Let $\hat{\vec{x}} = R^{-1}Q^\top \vec{b}$. Since $A = QR$, we have $A\hat{\vec{x}} = QRR^{-1}Q^\top \vec{b} = QQ^\top \vec{b}$. By Theorem 12, the columns of Q form an orthonormal basis for $\text{Col } A$, so $QQ^\top \vec{b}$ is the orthogonal projection of $\vec{b}$ onto $\text{Col } A$. Thus, $A\hat{\vec{x}} = \text{proj}_{\text{Col } A} \vec{b}$, which shows that $\hat{\vec{x}}$ is a least-squares solution. Uniqueness follows from Theorem 14.

6.7. Inner Product Spaces

Definition: Inner Product on a Vector Space

An inner product on a vector space V is a function that, to each pair of vectors $\vec{u}$ and $\vec{v}$ in V , associates a real number $\langle \vec{u}, \vec{v} \rangle$ and satisfies the following axioms for all $\vec{u}, \vec{v}, \vec{w}$ in V and all scalars c :

- $\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle$ (symmetry)
- $\langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$ (additivity)
- $\langle c\vec{u}, \vec{v} \rangle = c\langle \vec{u}, \vec{v} \rangle$ (homogeneity)
- $\langle \vec{u}, \vec{u} \rangle \geq 0$, and $\langle \vec{u}, \vec{u} \rangle = 0$ if and only if $\vec{u} = \vec{0}$ (positive definiteness)

A vector space with an inner product is an inner product space.

Theorem 16: The Cauchy-Schwarz Inequality

For all $\vec{u}, \vec{v}$ in an inner product space V :

$$|\langle \vec{u}, \vec{v} \rangle| \leq \|\vec{u}\| \|\vec{v}\|$$

where $\|\vec{u}\| = \sqrt{\langle \vec{u}, \vec{u} \rangle}$ (norm induced by the inner product).

Theorem 17: The Triangle Inequality

For all $\vec{u}, \vec{v}$ in an inner product space V :

$$\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$$

Generalized Orthogonalization

Given a basis $\{\vec{x}_1 \cdots \vec{x}_p\}$ for a nonzero subspace W of $\mathbb{R}^n$, define:

$$\begin{aligned}\vec{v}_1 &= \vec{x}_1 \\ \vec{v}_2 &= \vec{x}_2 - \frac{\langle \vec{x}_2, \vec{v}_1 \rangle}{\langle \vec{v}_1, \vec{v}_1 \rangle} \vec{v}_1 \\ &\vdots \\ \vec{v}_p &= \vec{x}_p - \sum_{i=1}^{p-1} \frac{\langle \vec{x}_p, \vec{v}_i \rangle}{\langle \vec{v}_i, \vec{v}_i \rangle} \vec{v}_i\end{aligned}$$

Then $\{\vec{v}_1 \cdots \vec{v}_p\}$ is an orthogonal basis for W .

In addition, $\text{Span}\{\vec{x}_1 \cdots \vec{x}_p\} = \text{Span}\{\vec{v}_1 \cdots \vec{v}_p\}$.